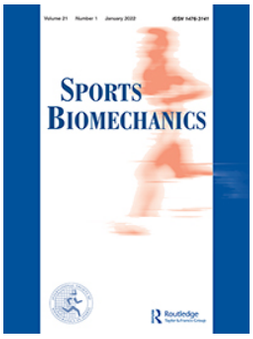

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Rating of perceived exertion and velocity loss as variables for controlling the level of effort in the bench press exercise

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ABSTRACT

There is a growing interest in the analysis of different methods for monitor fatigue during resistance training sessions. This study aimed to (1) analyse the relationships between the percentage of performed repetitions with respect to the maximum possible number (%REP), RPE and magnitude of velocity loss (VL), and (2) examine whether a multiple regression analysis with the RPE and VL as predictor variables could improve the goodness of fit to predict %REP in the bench press exercise performed in a Smith machine. Seven men performed a repetition maximum test, on 3 separate testing sessions, against 3 different absolute loads based on a target mean velocity (MV) according to an individual load-velocity profile (≈1.00, ≈0.70, and ≈0.50 m/s). MV, VL, %REP and RPE were collected and used for analysis. Based upon quadratic polynomial regression analysis strong relationships were reported between the RPE and %REP (r^2 = 0.89 and SEE = 9.85%) and between the VL and %REP (r^2 = 0.91 and SEE = 9.85%). Multiple regression analysis with the RPE and VL as predictor variables improved the goodness of fit (r^2 = 0.94 and SEE = 7.18%) of the model to predict %REP. These results suggest that both RPE and VL are useful variables to accurately estimate %REP in the bench press exercise.

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Velocity-based training; subjective scale; level of effort; exercise prescription

Introduction

Resistance training has been shown to be a key factor for improving health, body aesthetics and athletic performance (Balshaw et al., 2017; Folland & Williams, 2007; Suchomel, Nimphius, & Stone, 2016). However, coaches and sport professionals require an adequate control of the different training variables (e.g., intensity, volume, movement velocity, rest periods, etc.) in order to design an effective resistance training program (Kraemer & Ratamess, 2004). Among these training variables, the training intensity and volume seems to be the most important factors impacting the athlete's neuromuscular adaptations (Campos et al., 2002; Gonzalez-Badillo, Yanez-Garcia, Mora-Custodio, & Rodriguez-Rosell, 2017; Munoz-Lopez, Marchante, Cano-Ruiz, Chicharro, & Balsalobre-Fernandez, 2017).

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Training intensity has traditionally been based on percentage of the one-repetition maximum (%1RM) or on the maximum number of repetitions that can be performed against a submaximal load (Kraemer & Ratamess, 2004; Sakamoto, Sinclair, & Naito, 2016). More recently, is has been suggested that movement velocity may be an appropriate way to control training intensity (González-Badillo & Sánchez-Medina, 2010; Jidovtseff, Harris, Crielaard, & Cronin, 2011; Pareja-Blanco et al., 2017). Previous studies have observed a strong and stable relationship between movement velocity and the relative load (%1RM) in a variety of resistance training exercises, mostly performed in Smith machines (Conceição et al., 2016; García-Ramos, Pestaña-Melero, Pérez-Castilla, Rojas, & Haff, 2018a; Munoz-Lopez et al., 2017; Pérez-Castilla, García-Ramos, Padial, Morales-Artacho, & Feriche, 2017). Velocity-based training has therefore gained popularity in the field of strength and conditioning due to the proliferation of and advancements in sports technology (i.e. linear transducers, accelerometers, or mobile applications) that allow for the accurate measurement of movement velocity (Balsalobre-Fernandez et al., 2017; Banyard, Nosaka, Sato, & Haff, 2017; Pérez-Castilla, Piepoli, Delgado-García, Garrido-Blanca, & García-Ramos, 2019). However, many of these devices are not able to be practically used within the strength and conditioning environment and are often not affordable, which limits their ability to be used by many strength and conditioning professionals (Pérez-Castilla et al., 2019).

An alternative approach to monitor resistance training intensity based on the psycho-physiological response of the organism is the 'rating of perceived exertion' (RPE) (Day, McGuigan, Brice, & Foster, 2004; Eston & Evans, 2009; Gearhart, Lagally, Riechman, Andrews, & Robertson, 2009). According to this, the psychological/environmental information provokes a subjective perception for a given stimulus (Borg, 1990). The scales commonly used to collect RPE are the Borg (Borg, 1990) or OMNI-RES (Robertson et al., 2003) scales. The RPE has been shown to be a useful tool for predicting the relative load (%1RM) (Day et al., 2004; Eston & Evans, 2009; Gearhart et al., 2009). However, the relationship between the RPE and resistance training intensity may be questionated when performing sets to voiluture failure due to the induced fatigue (Vasquez et al., 2013). Previous investigations have found high relationship between the RPE, %RM, and MV (Bautista, Chirosa, Robinson, Chirosa, & Martinez, 2016; Naclerio & Larumbe-Zabala, 2016). Also, a novel scale in which RPE was based on the number of 'repetitions in reserve' (i.e., number of repetitions left in reserve before reaching muscular failure in a training set; RIR) has been developed (Zourdos et al., 2016). More importantly, a recent study by Balsalobre-Fernández, Muñoz-López, Marchante, and García-Ramos (2018) observed that combining RPE, RIR and movement velocity improves the capabilities of traditional load-velocity relationships to predict relative loads (%RM). Therefore, it is well documented in scientific literature the feasibility of RPE scales to prescribe and monitor training load according to athlete's subjective perception.

Of special note, is that training intensity in isolation is an inadequate predictor of training adaptation if training volume is not taken into account (Gonzalez-Badillo et al., 2017). It has been widely argued in the scientific literature that training to muscular failure (i.e., performing the maximum number of repetitions possible with a given relative load) is unnecessary for maximising athletic performance (Davies, Orr, Halaki, & Hackett, 2016; Izquierdo et al., 2006). Velocity loss (VL) has been recently proposed

as a precise and non-invasive variable to monitor the actual degree or level of effort incurred during resistance training (Sánchez-Medina & González-Badillo, 2011). Previous studies have reported a strong relationship between the relative VL in a training set and the percentage of performed repetitions with respect to the maximum possible number that can be completed (%REP) in the bench press and pull-up exercises in a smith machine (Gonzalez-Badillo et al., 2017; Sanchez-Moreno, Rodriguez-Rosell, Pareja-Blanco, Mora-Custodio, & Gonzalez-Badillo, 2017). By monitoring repetition velocity can be accurately estimated with the %RPE (or RIR) when a given percentage of VL is achieved during a training set. However, no known studies have analysed whether the subjective RPE scale is able to estimate the %RPE with similar precision to that seen with the VL method. It also remains unknown whether incorporating the RPE along with VL has additional predication capabilities to monitoring %RPE.

In line with the aforementioned considerations, the objectives of the present study were to (1) analyse the relationships between the %REP, RPE and VL, and (2) examine whether a multiple regression analysis with the RPE and VL as predictor variables could improve the precision of the prediction of the %REP in the bench press exercise performed in a Smith machine. We hypothesised that (1) no difference would be observed between the RPE and VL in the precision of the prediction of %REP, (2) a more precise estimation of the %RPE would be obtained using RPE and VL as predictor variables.

Methods

Participants

The sample size was determined based on previous studies (Sreckovic et al., 2015; Zivkovic, Djuric, Cuk, Suzovic, & Jaric, 2017), which calculated that a sample size between 3 and 12 participants was needed to detect differences on mechanical parameters observed from the loaded bench press executions with a significant level of $p < 0.005$ and desired statistical at 0.8. Therefore, we recruited 7 physically active men (age: 22.9 ± 1.9 years; body mass: 77.8 ± 11.7 kg; height: 1.76 ± 0.68 m) volunteered to participate in this study. Participants had at least 1 year of resistance training experience and a bench press 1RM of 83.6 ± 16.4 kg (1.08 ± 0.20 kg/kg body mass). No physical limitations, health problems or musculoskeletal injuries that could compromise tested performance were reported. None of the participants were taking drugs, medications or other substances that could alter their physical performance. Participants were informed of the study procedures and signed a written informed consent form prior to initiating the study. The study protocol adhered to the tenets of the Declaration of Helsinki and was approved by the Institutional Review Board at the University of Granada.

Experimental design

A cross-sectional study was designed to examine the RPE-%REP-VL relationships in the bench press exercise. Participant undertook 3 testing session, at the same time of

the day for each participant, separated by 72–96 hours of rest between them. In each testing session, participants completed a repetition maximum test against an external load based on a target mean velocity (MV) according to previously established individual load-velocity profile (≈1.00, ≈0.70, and ≈0.50 m/s). The individual load-velocity relationship was was determined based upon the completion of a standardized incremental load testing until a 1RM has been achieved (García-Ramos et al., 2018a). During the repeptition maximum test, the RPE scale was inmediately shown to the participants and they were asked to verbally rating the RPE value during each analysis interval previously established (see above for further details). The main variables analysed were: 1) mean velocity (MV) of each repetition; 2) MV of the fastest repetition (MV_{BEST}) during the repetition maximum test; 3) MV of the last repetition (MV_{LAST}) during the repetition maximum test; 4) velocity loss (VL) over each analysis interval calculated as $100^*(MV_{BEST}_MV/MV_{BEST})$; 5) percentage of repetitions completed (%REP) with respect to the maximum possible number over each analysis interval; and 6) rating of perceived exertion (RPE) verbally reported over each analysis interval.

Testing procedures

Familiarisation and initial measures

In the 2 weeks prior to the study, participants performed 2 familiarisation sessions, separated by 7 days, to ensure a proper technique during the bench press exercise and familiarise participants with Borg's RPE (1–10) scale (Borg, 1990). In the first familiarisation session, the anthropometric measures (height and body mass) were taken and then participants carried out a standard incremental loading test to determine the individual load-velocity profile (García-Ramos, Pestaña-Melero, Pérez-Castilla, Rojas, & Haff, 2018b). In the second familiarisation session, participants performed several warm-up sets of 3–8 repetitions with different external loads and, subsequently, had to verbally report the RPE value after each set. Thereafter, they carried out a repetition maximum test against an external load based on a target MV of ≈0.70 m/s according to the previously established individual load-velocity profile in order to experience a maximum effort (i.e., RPE 10).

Load-velocity profile determination

Before the commencement of the load-velocity test, participants performed a standardised warm-up consisting of 5 min of joint mobility, followed by 2 sets of 10 and 8 repetitions (interspaced with 2 min rest) with an external load of 20 and 30 kg respectively. After warming up, the individual load-velocity relationship was determined by means of a standard incremental loading test performed in a Smith machine (Technogym, Barcelona, Spain) (García-Ramos et al., 2018a). The initial external load of the incremental loading test was set at 20 kg. Thereafter, the load was progressively increased in 10 kg increments until the MV was lower than 0.50 m/s. From that moment, the load was progressively increased in steps of 5 to 2.5 kg until the 1RM strength was achieved. For the lighter loads (MV ≥ 0.80 m/s), 3 repetitions were performed at each load, 2 for medium loads (0.80 m/s > MV ≥ 0.5 m/s), and only 1 for the heavier loads (MV < 0.5 m/s). Participants were verbally asked to provide a RPE value after each load lifted in order to familiarise them with this RPE scale.

The bench press technique involved the participants laying supine on flat bench, with their feet fully supported on the floor and their hands placed on the bar with a self-selected grip-width, which was slightly wider than shoulder width. The self-selected grip width was individually measured and kept constant throughout all testing sessions. The position on the bench was individually adjusted with the vertical projection of the bar corresponding with the participant's mid line of the chest. Participants were required to lower the bar in a controlled manner until contacting their chest. After maintaining a momentary pause for ≈1–1.5 s, they performed a purely concentric action at the maximum possible velocity. The momentary pause between phases was imposed in order to minimise the effect of the stretch-shortening cycle and the contribution of elastic energy to the movement as proposed previously (Pareja-Blanco et al., 2017).

Repetition maximum test

Participants completed a standardised warm-up consisted of 5 min of joint mobility, followed by 4 sets of 10 repetitions against 20 kg, 5 repetitions against 30 kg, 5 repetitions against 40 kg and 2 repetitions against 50 kg (interspaced with 2 min rest). If the participant reported a higher 1RM than 80 kg, an additional set of 2 repetitions against 60 kg was completed. Afterward, participants performed a repetition maximum test based on completing the maximum number of possible repetitions with a submaximal load previously determined according to the individual load-velocity profile. Specifically, the absolute loads (kg) were established based on a target MV reached at the first 1–3 repetitions (usually the fastest) during each repetition maximum test: load 1 = ≈1.00 m/s, load 2 = ≈0.70 m/s, and load 3 = ≈0.50 m/s. The repetition maximum test ended when participants were unable to perform a repetition with a proper technique or when a completed range of motion was not reached (Pérez-Castilla, Suzovic, Domanovic, Fernandes, & García-Ramos, 2019). A pause of 2 s. was imposed between the eccentric and concentric phases of each repetition. During each repetition maximum test, the RPE value was inmediately requested from the participant in pre-established analysis intervals: every 5, 3 and 2 repetitions during the load 1 (≈1.00 m/s), 2 (≈0.70 m/s), and 3 (≈0.50 m/s), respectively.

Measurement equipment and data analysis

Anthropometric variables (height and body mass) were measured with a digital stadiometer with scale (SECA 220, SECA, Germany). A Smith machine (Technogym, Barcelona, Spain) was used for all tests. An isoinertial dynamometer (ChronojumpTH Boscosystem, Barcelona, Spain), which consists of a cable-extension linear position transducer attached to the bar interfaced with a personal computer at a sampling rate of 1,000 Hz, was used for mechanical measurements. Specifically MV, which is determined by the differentiation of the displacement data with respect to time, was automatically calculated by the custom software v.1.8.1–54. Reliability and validity of this device has been previously reported else where (Pérez-Castilla et al., 2019). A visual presentation of Borg's RPE scale (1–10) was used throughout testing.

Statistical analyses

Descriptive data are presented as means, standard deviations (SD) and coefficients of variation (CV) with the corresponding 95% confidence intervals (CI). The distribution

of the variables was analysed by the Kolmogorov-Smirnov tests. The non-parametric Kruskal-Wallis *H* tests were used to compare dependent variables between the repetition maximum tests. Mann-Whitney *U* tests were used to identify the source of any significant difference. Quadratic polynomial regression models were used to examine the relationship between %REP, RPE and VL. The goodness of fit was assessed by the Pearson's multivariate coefficient of determination (r^2) and the standard error of the estimate (SEE). Multiple regression analysis was conducted to estimate the %REP using RPE and VL as predictor variables. The validity of the prediction equations was examined through the Pearson's correlation coefficient (r) with 95% CI via bootstrapping; and Bland-Altman plots were used to identify potential systematic bias. The criteria for interpreting the magnitude of the r coefficients were: trivial (0.00–0.09), small (0.10–0.29), moderate (0.30–0.49), large (0.50–0.69), very large (0.70–0.89), nearly perfect (0.90–0.99) and perfect (1.00) (Hopkins, Marshall, Batterham, & Hanin, 2009). Heteroscedasticity of error was defined as an r^2 of 0.1 (Atkinson & Nevill, 1998). The level of significance was established at $p \leq 0.05$. Statistical analyses were performed using the software package SPSS (IBM SPSS version 23.0, Chicago, IL, USA).

Results

Descriptive data

The Kruskal-Wallis *H* test only revealed significant differences for the variables VL (Chi^2: 7.85) and repetitions (Chi^2: 17.4). The pairwise comparisons are reported in Table 1.

%REP-RPE and %REP-VL relationships

Quadratic polynomial regressions for the relationship between the RPE-%REP and between the VL-%REP are presented in Figure 1. Multiple regression analysis for the prediction of %REP (dependent variable) by RPE and VL (independent variables) obtained from the whole dataset (n = 140) reported a high goodness of fit (r^2 = 0.94; SEE = 7.18%; $p < 0.001$) following the next equation: %REP = 0.76VL + 4.27RPE—2.07.

Validity of the equations to predict %REP

When analysing the whole dataset (n = 140), nearly perfect correlations (r range = 0.94–0.95; SEE range = 8.52–9.24%) between the actual and predicted %REP values were observed for both the RPE and VL relationship (Figure 2). Similarly, the Bland-Altman plots showed trivial systematic bias (%REP predicted by RPE = −0.00%; %REP predicted by VL = −0.05%) and moderate random errors (%REP predicted by RPE = 9.78%; %REP predicted by VL = 8.89%) (Figure 3). No heteroscedasticity of the errors was observed (r^2 range = 0.02–0.03).

Table 1. Descriptive variables obtained during the lift-to-failure tests in the bench press exercise.

	Load 1 (≈ 1.00m/s)		Load 2 (≈ 0.70m/s)		Load (≈ 0.50m/s)	
Variables	Mean ± SD (range)	CV (95% CI)	Mean ± SD (range)	CV (95% CI)	Mean ± SD (range)	CV (95% CI)
MV_{BEST} (m/s)	1.00 ± 0.05 (0.91–1.06)‡	4.7 (0.9–1.0)	0.71 ± 0.03 (0.68–0.76)*	3.9 (0.6–0.7)	0.51 ± 0.03 (0.47–0.55)*‡	6.0 (0.4–0.5)
MV_{LAST} (m/s)	0.20 ± 0.08 (0.11–0.31)	41.8 (0.1–0.3)	0.15 ± 0.03 (0.10–0.19)	19.6 (0.1–0.2)	0.15 ± 0.03 (0.11–0.18)	18.5 (0.1–0.2)
VL (%)	80.20 ± 9.0 (67.1–88.9)	11.2 (71.9–88.5)	79.0 ± 4.1 (75.1–85.4)	5.2 (75.2–82.8)	69.3 ± 6.5 (61.7–78.3)*‡	9.4 (63.3–75.3)
Load (kg)	36.4 ± 5.8 (30.0–45.0)‡	15.8 (31.1–41.7)	53.2 ± 8.6 (40.0–65.0)*	16.2 (45.2–61.2)	64.6 ± 9.9 (52.5–77.5)*	15.4 (55.5–73.8)
Repetitions	41.4 ± 15.8 (24.0–74.0)‡	38.2 (26.8–56.1)	19.3 ± 4.1 (13.0–25.0)*	21.1 (15.5–23.1)	9.6 ± 1.4 (8–12)*‡	14.6 (8.3–10.9)

SD, standard deviation; CV, coefficient of variation; 95% CI, 95% confidence interval; MV_{BEST}, mean velocity of the fastest repetition during the lift-to-failure test; MV_{LAST}, mean velocity of the last repetition during the lift-to-failure test; VL, velocity loss over each analysis interval. *, significant differences with respect to load 1; ‡, significant differences with respect to load 2; ǂ significant differences with respect to load 3 ($p < 0.05$; Mann-Whitney U test).

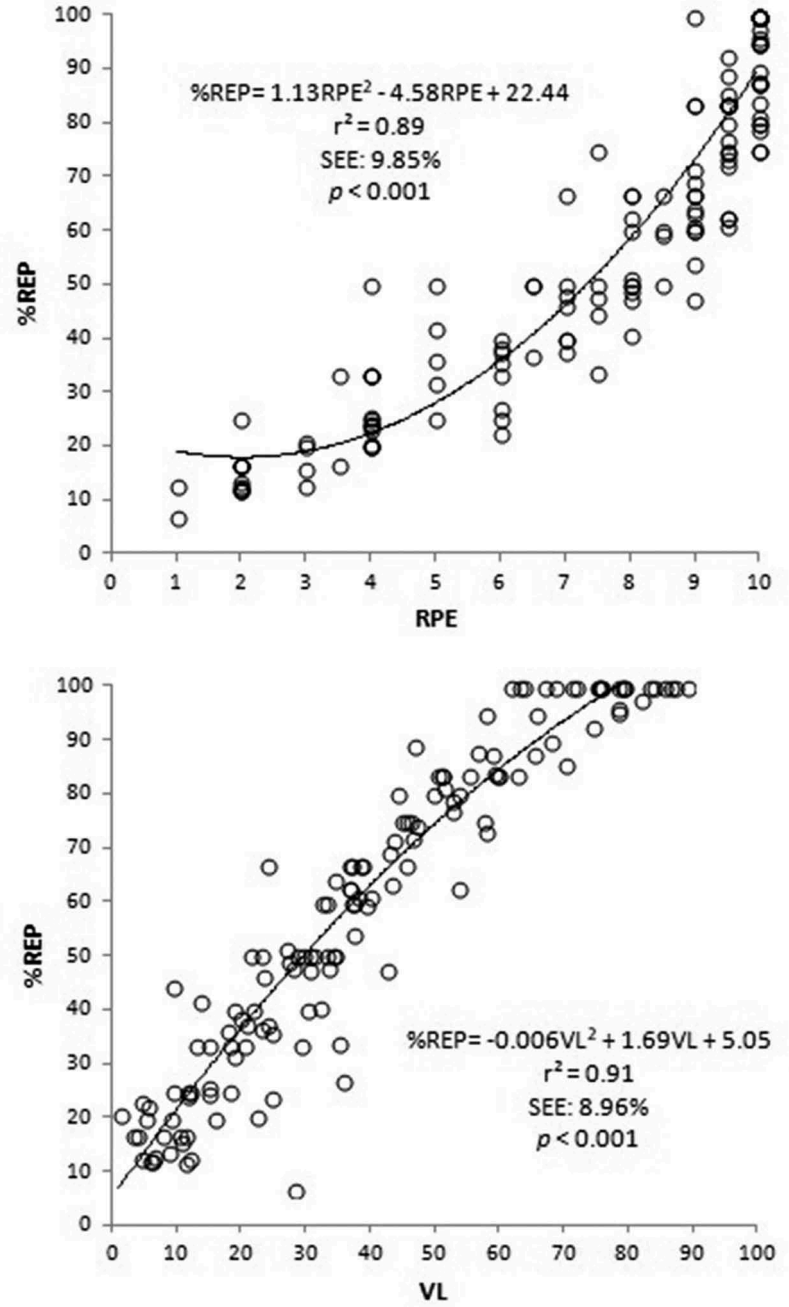

Figure 1. Quadratic polynomial regressions models obtained for the relationships between the RPE and the percentage of performed repetitions with respect to the maximum possible number (%REP; upper panel) and the magnitude of velocity loss and %RPE (lower panel). r^2, Pearson's multivariate coefficient of determination; SEE, standard error of the estimate, *p*, *p*-value.

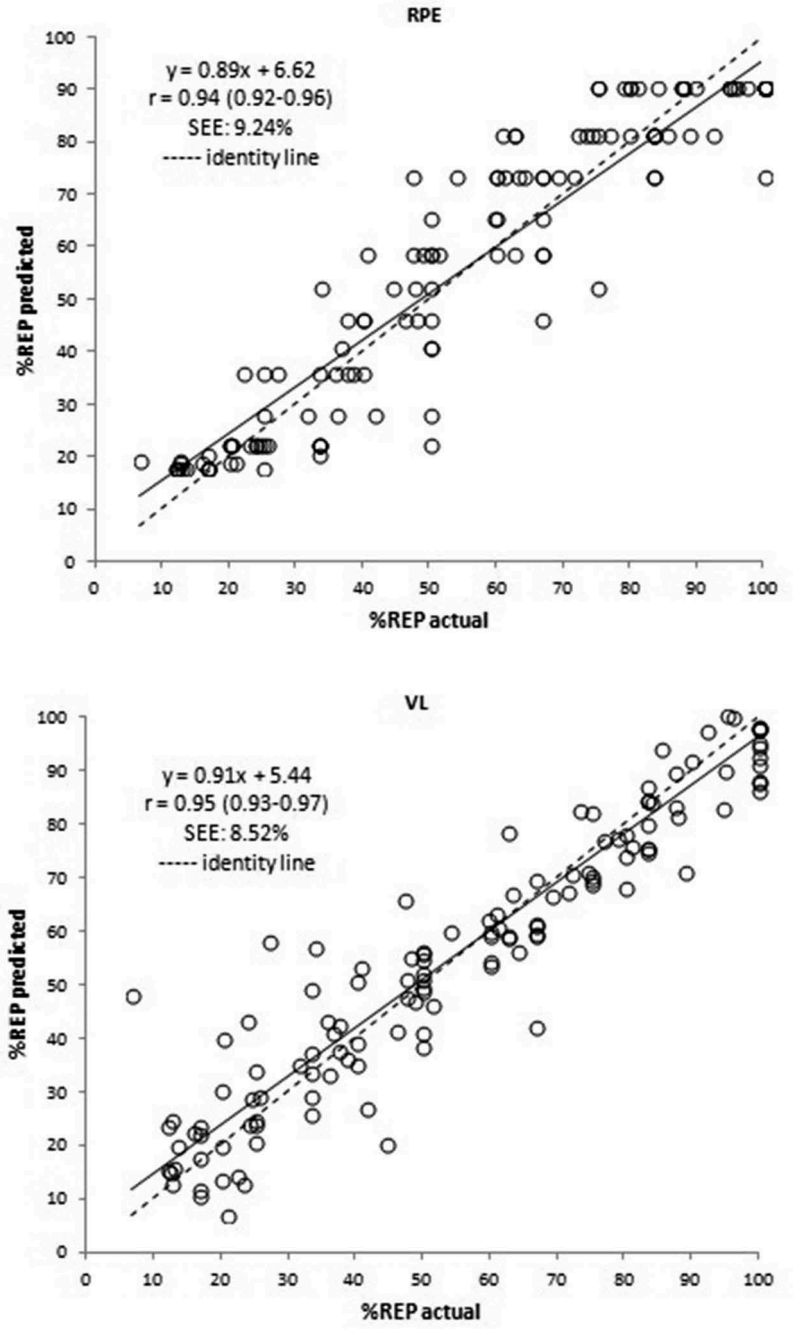

Figure 2. Relationship between the actual and predicted values of percentage of performed repetitions with respect to the maximum possible number (%REP) obtained from RPE (upper panel) and the magnitude of velocity loss (VL; lower panel). *r*, Pearson's correlation coefficient (*r*) with 95% confidence intervals; SEE, standard error of the estimate.

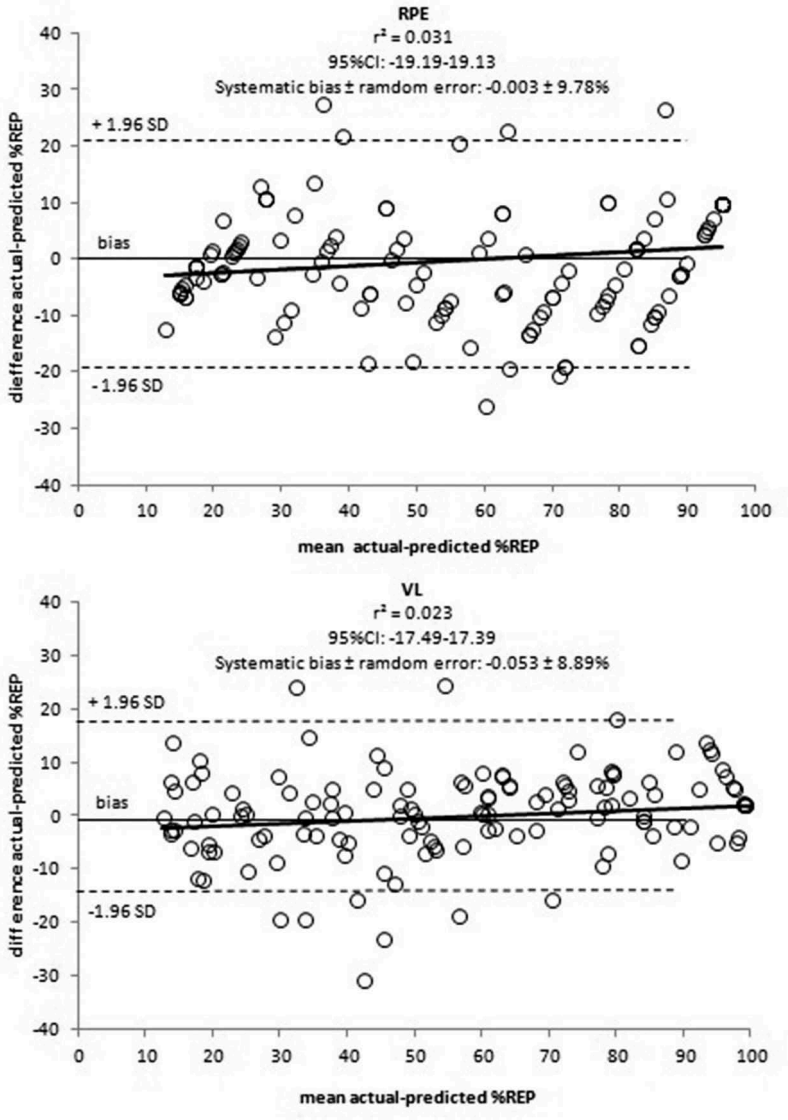

Figure 3. Bland-Altman plots showing differences between the actual and predicted values of percentage of performed repetitions with respect to the maximum possible number (%REP) obtained from RPE (upper panel) and the magnitude of velocity loss (VL; lower panel). Each plot depicts the averaged differences and 95% limits of agreement (± 1.96 SD; dashed lines), along with the regression line (solid line). The systematic bias ± random error together with strength of the relationship (r^2) and its 95% confidence interval (CI) is depicted in each plot.

Discussion and implications

This study was designed to analyse the relationships between the RPE, VL and %RPE in the bench press exercise performed with a Smith machine following 3 different repetition maximum tests. Supporting our first hypothesis, the relationships between %REP-RPE and %REP-VL were both very high as revealed by the regression analysis.

Our second hypothesis was also confirmed since incorporating the RPE together with VL slightly improved the goodness of fit to predict %RPE. Taken together, these findings highlight that both RPE and VL can be useful variables when attemting to estimate %REP in the bench press exercise performed with a Smith machine. Using this approach, strength and conditioning professionals could objectively quantify the level of effort incurred during the bench press exercise performed in a Smith machine.

Previous studies have confirmed the feasibility of RPE for monitoring resistance training due to strong relationships observed between the RPE, relative load (%1RM) and MV (Bautista et al., 2016; Naclerio & Larumbe-Zabala, 2016). However, this study is the first to our knowledge that has evaluated the precision of the RPE as a tool for predicting %REP during resistance exercise. Our results show a strong relationship between the RPE and % REP in the bench press exercise. In addition, nearly perfect correlations and trivial systematic bias was observed between actual and predicted %REP values through RPE. Therefore, this study adds evidence that supports the use of RPE scale as a practical tool to objectively quantify the level of effort incurred during resistance training. However, it should be keep in mind that fatigue may be a potencial confounding factor when attemting to utilise RPE in resistance training (Vasquez et al., 2013).

Supporting our first hypothesis, a strong relationship was observed between the VL and %RPE, being the goodness of fit comparable to the RPE-%RPE relationship. Similar to the RPE-%REP relationship, nearly perfect correlation and trivial systematic bias were observed between actual and predicted %REP values through VL. The present results agree with those previously presented for the bench press and pull-up exercises (Gonzalez-Badillo et al., 2017; Sanchez-Moreno et al., 2017). Our results revealed a goodness of fit somewhat higher (r^2 = 0.91 *vs*. 0.88 and SEE = 8.96 *vs*. 9.44%) than those observed during the pull-up exercise (Sanchez-Moreno et al., 2017). However, when comparing our results with those presented for the bench press exercise by Gonzalez-Badillo et al. (2017), a lower r^2 (0.91 *vs*. 0.97) and higher SEE (8.96% *vs*. 5.40%) can be observed. These differences could be attributed to three methodological differences: (1) the number of relative intensities used in modelling (3 *vs*. 8), (2) the analysis procedure of the VL values (pre-established analysis of repetition intervals *vs*. pre-fixed VL thresholds) and (3) the type of velocity used in the analysis (mean concentric velocity in our study *vs*. mean concentric propulsive velocity in Gonzalez-Badillo et al. (2017)). Therefore, our results also corroborate that monitoring repetition velocity can be useful approach to provide an accurate estimation of the %RPE during resistance training. Of special note is that the current findings were obtained during the bench press exercise performed in a Smith machine and, therefore, the ecological validity of measures can be limitated in comparisons with the free-weight bench press exercise. However, it has been previously demonstrated that the incorporation of machine-based testing may provide more reliable velocity measurements when compared with free-weight bench exercises (Miller et al., 2018).

To the best of our knowledge, this is the first study that has examined whether incorporating the RPE along with VL as predictor variables in modelling could improve the goodness of fit to predict %REP. Supporting our second hypothesis, the multiple regression analysis shows that utilising both variables (VL and RPE) could increase the explained variance and reduce the SEE compared to the quadratic regressions models using VL or RPE alone. The present results agreed with the study of Balsalobre-Fern

ández et al. (2018) where the combination of RPE and MV showed higher precision for estimating relative load than the traditional load-velocity relationship. These results indicate that the combination of RPE and VL could be useful for increasing the precision of the effort monitoring process.

The main limitation of the current study was the relatively small sample size, and the fact that the participants presented different training backgrounds, which may be a potential confounding factor. Note that previous studies have highlighted that more experienced lifters and those with previous experience using RPE scales could report more accurate estimation of training intensity (i.e., %RM or MV) than novice inexperienced lifters (Helms et al., 2017; Zourdos et al., 2016). Futures studies should examine the possible effect of training backgrounds on predication capabilities of the RPE-%REP relationship. It also needs to be considered that the results in our study were obtained during a paused bench-press performed in a Smith machine, and they should not be generalized to other training conditions. Although the load-velocity relationship seems not to dependent on the bench press execution mode (i.e., free-weight or Smith machine) (Loturco et al., 2017), futures studies should examine whether our findings could be applicable to the free-weight bench press, other free-weight resistance training exercises (e.g., back-squat, deadlift, etc.) as well as other execution techniques (e.g., the eccentric-concentric bench press variant).

Conclusions

This study confirmed the feasibility of the RPE and VL variables to accurately estimate %REP in each exercise set due to the strong relationships observed between the RPE, VL and %RPE during the bench press exercise performed with a Smith machine. However, since the goodness of fit is improved by incorporating RPE or VL as predictor variables, the subjective scale should be considered in conjunction with the velocity measurements to obtain a more precise estimation of the %RPE through the generalised group equation presented in the current study. The results of this study contribute by improving the knowledge regarding the resources that can be used to prescribe, monitor and quantify Smith machine becnch press training . Specifically, our results are of great practical application because they allow one to objectively quantify the level of effort during each training set in the Smith machine bench press exercise.

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Disclosure statement

No potential conflict of interest was reported by the authors.

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